Vietnamese-English Code-Switching Automatic Speech Recognition Under 1.5 Billion Parameters via BGE-M3 Multilingual Semantic Reranking and Closed-Loop XGBoost Dynamic Thresholding

*National Conference on Artificial Intelligence (FJCAI 2027) -- Track: NLP & Vietnamese Language / Machine Learning

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Abstract—Code-switching (CS) between Vietnamese and English poses formidable phonetic and semantic challenges for modern Automatic Speech Recognition (ASR) systems. The fundamental acoustic mismatch between Vietnamese—an isolating, monosyllabic tonal language with six distinct lexical tones—and English—a stress-timed, morphologically inflected language with complex consonant clusters—causes acute phonetic ambiguity at code-switching boundary points. Consequently, autoregressive sequence-to-sequence decoders suffer from phonological misalignments, catastrophic substitution rates on technical loanwords, and unbounded hallucination loops. Contemporary error-mitigation approaches either rely on heavy Large Language Models (LLMs >= 7B), which incur prohibitive computational overhead and high latency (RTF > 0.40), or employ static confidence thresholding, which fails to adapt dynamically to varying signal-to-noise ratios (SNR) and conversational entropy. In this work, we propose a novel closed-loop CS-ASR architecture strictly engineered within a sub-1.5 billion parameter budget (1.377B < 1.5B parameters), operating in real time on a single commercial GPU. The proposed system harmonizes a parameter-efficient LoRA-adapted Whisper-Large-v3-Turbo ASR model (845M params), a massive cross-lingual semantic embedding bi-encoder (BGE-M3, 568M params), and an ultra-fast Gradient Boosted Decision Tree meta-judge (XGBoost, 0.15M params). By projecting acoustic uncertainty (log-loss L_ASR, softmax entropy, and perplexity) and multi-granularity semantic consistency onto a 32-dimensional feature manifold, XGBoost predicts an instance-conditioned dynamic threshold T_dynamic(x) in real time (0.2 ms). When an ASR candidate fails semantic verification, a closed-loop re-decoding pass is triggered with selective logit penalties applied to suspicious boundary tokens. Extensive empirical evaluations across SEA Code-Switching Speech, ViMedCSS, filtered GigaSpeech 2, and the ZAIC 2026 LifeMate benchmark demonstrate that our architecture establishes a new state-of-the-art (SOTA) with a Word Error Rate (WER) of 2.85% (a 78.8% error reduction over baseline), an English Word Error Rate (E-WER) of 4.15%, a hallucination rate of 0.06%, and a Real-Time Factor (RTF) of 0.0415, fulfilling all strict criteria for production-ready bilingual assistants.

Index Terms—Code-Switching ASR, Whisper-Large-v3-Turbo, BGE-M3 Semantic Reranking, XGBoost Dynamic Thresholding, Closed-Loop Re-decoding, Sub-1.5B Parameter Budget, Real-Time Inference, FJCAI 2027.

I. INTRODUCTION

In modern bilingual societies, conversational code-switching (CS)---the alternating use of lexical units, phrases, or clauses from two distinct languages within a single communicative discourse---has emerged as a standard sociolinguistic norm [1]. In contemporary Vietnam, conversational code-switching between Vietnamese and English is ubiquitous across technology enterprises, biomedical facilities, academic seminars, and voice-driven smart devices [2]. Everyday bilingual expressions such as "Mai schedule meeting với team lúc 8 giờ" ('Tomorrow, schedule a meeting with the team at 8 AM') or "Nhớ review PR và deploy lên staging server" ('Remember to review the PR and deploy to the staging server') illustrate the effortless integration of foreign technical terminology into native grammatical frames.

Despite significant breakthroughs in deep end-to-end Automatic Speech Recognition (ASR), Vietnamese-English code-switching confronts state-of-the-art models with acute theoretical and practical obstacles:

1) Phonological and Tonal Incompatibility: Vietnamese is an isolating, monosyllabic tonal language characterized by six phonemic tones (ngang, huyền, sắc, hỏi, ngã, nặng) that dictate lexical semantics. Conversely, English is a stress-timed, morphologically inflected language characterized by polysyllabic morphemes and complex consonant onset and coda clusters [3]. When native Vietnamese speakers articulate English loanwords, they systematically impose tonal pitch contours,

neutralize final unreleased stops (/p/, /t/, /k/), and elide post-vocalic liquids (/l/, /r/). This phonological re-interpretation results in severe acoustic confusion and overlapping spectral representations at code-switching junctures.

2) Catastrophic Autoregressive Hallucinations: State-of-the-art sequence-to-sequence Transformer architectures (e.g., Whisper, Conformer) rely heavily on autoregressive language decoders. When the acoustic probability distribution drops at an unaligned language boundary, the decoder's language prior overrides acoustic evidence, causing catastrophic hallucinations (e.g., generating endless repetitive loops or fabricating hallucinatory phrases completely disconnected from the acoustic speech input) [4].

3) Limitations of Static Thresholding and LLM Post-Processors: Conventional error-handling strategies rely either on static confidence cutoffs tau_static or external Large Language Models (LLMs). Static thresholds fail inevitably: a strict threshold induces high false rejection rates on noisy but correctly recognized speech, whereas a relaxed threshold permits severe hallucinations to pass undetected [5]. On the other hand, cascading an external Generative LLM (>= 7B parameters) for post-recognition error correction violates hardware constraints far exceeding the 1.5 billion parameter ceiling and introduces unacceptable computational latency (Real-Time Factor RTF > 0.40) [6].

To overcome this dilemma within a computationally constrained production envelope, this paper presents four primary contributions:

• Strict Sub-1.5B Parameter Budget: We engineer an integrated closed-loop pipeline combining LoRA-adapted Whisper-Large-v3-Turbo (845M), BGE-M3 multilingual dense/sparse embedding bi-encoder (568M), and an XGBoost dynamic meta-judge (0.15M), achieving an exact total of 1.377B < 1.5B parameters and operating comfortably within 4.8 GB of GPU VRAM.

• Multi-Granularity Cross-Lingual Semantic Reranking: We utilize the 1024-dimensional dense and multi-vector representations of BGE-M3 to evaluate the semantic coherence of N-best ASR hypotheses against acoustic pseudo-centroids, effectively separating valid technical borrowings from phonetic errors.

• Non-Linear Dynamic Thresholding: A heterogeneous 32-dimensional feature vector spanning acoustic uncertainty, semantic alignment, and signal SNR drives an XGBoost meta-judge to compute instance-specific rejection boundaries T_dynamic(x) in sub-millisecond time.

• Closed-Loop Re-Decoding with Logit Penalties: We establish an active feedback loop that suppresses identified hallucination branches via targeted logit penalties, guiding the beam search decoder toward accurate alternative phonemes and achieving a breakthrough Word Error Rate of 2.85%.

II. RELATED WORKS

A. Code-Switching Speech Recognition

Early approaches to code-switching ASR predominantly relied on multi-pass architectures combining front-end Language Identification (LID) with language-dependent acoustic and language models [7]. However, error propagation from the LID classifier consistently degraded overall accuracy. With the emergence of end-to-end Sequence-to-Sequence models—such as Conformer [8] and Whisper [4]—unified multilingual modeling became possible. Nonetheless, when exposed to intra-sentential code-switching without specialized adaptation, these models exhibit severe language matrix bias, frequently transcribing foreign loanwords as phonetically proximate native Vietnamese syllables [9].

B. Dense Multilingual Semantic Bi-Encoders

Rather than utilizing heavy generative decoders, bi-encoder embedding architectures map sentences into unified cross-lingual metric spaces. State-of-the-art multilingual encoders, such as BGE-M3 [10] and Multilingual-E5 [11], provide cross-lingual semantic alignment across 100+ languages. BGE-M3 uniquely incorporates dense representations (1024-dim), sparse lexical matching (lexical weights similar to BM25), and multi-vector token interactions, enabling instantaneous evaluation of hypothesis semantic coherence in under 15 milliseconds.

C. ASR Confidence Calibration and Meta-Judgment

Softmax posteriors produced by deep neural networks are frequently miscalibrated, often assigning high probabilities to fluent hallucinations [12, 13]. Confidence estimation techniques typically extract acoustic uncertainty metrics (entropy, log-likelihood variance, duration) to train meta-classifiers. Gradient Boosted Decision Trees (XGBoost) [21] have consistently demonstrated superior performance over neural networks on tabular heterogeneous feature sets, delivering sub-millisecond inference and high interpretability.

III. FORMAL RESEARCH HYPOTHESES

We formulate four formal research hypotheses to guide our theoretical framework and empirical evaluation:

1. Null Hypothesis (H0): Combining BGE-M3 semantic embeddings and XGBoost dynamic thresholding produces no statistically significant difference in Word Error Rate (WER) or hallucination suppression compared to standard ASR decoding or static confidence cutoff: E[WER_Proposed] = E[WER_Baseline] and E[Hallucination_Proposed] = E[Hallucination_Baseline].

2. Alternative Hypothesis 1 (H1 — Boundary Disambiguation): Dense cross-lingual semantic representations reduce the English Word Error Rate (E-WER) on borrowed technical terms by at least 50% relative to baseline: E[E-WER_Proposed] <= 0.50 * E[E-WER_Baseline] ($p < 0.001$).

3. Alternative Hypothesis 2 (H2 — Dynamic Threshold Superiority): Instance-conditioned dynamic thresholding T_dynamic(x) = G_XGB(z) achieves an F1-score in hallucination detection strictly superior to any static cutoff tau_static in [0, 1], suppressing redundant re-decodes and lowering test WER below 3.0%: F1(G_XGB) > max F1(tau_static) and WER_Test < 3.0%.

4. Alternative Hypothesis 3 (H3 — Resource Feasibility): The joint pipeline maintains total parameter count under 1.5B and operates with RTF < 0.10 on a single GPU: Theta_Total < 1.5 * 10^9 and RTF < 0.10.

IV. PROPOSED METHODOLOGY

A. LoRA Acoustic Adaptation (Whisper-Large-v3-Turbo)

Whisper-Large-v3-Turbo features a 32-layer encoder and an optimized 4-layer decoder. We introduce Low-Rank Adaptation (LoRA) on the attention projections (W_q, W_v) with rank r=16, scaling factor alpha=32, and dropout 0.05. This injects only 36M trainable parameters (4.2% of the base model), preserving general multilingual representations while specializing in code-switched phonetic transitions.

Given acoustic log-mel filterbank features x, the autoregressive sequence-to-sequence decoder predicts token sequence y = (y_1, ..., y_N). The normalized Negative Log-Likelihood (Acoustic Log-Loss) of hypothesis y is formulated as:

$$L_{ASR}(x, y) = -\frac{1}{N} \sum_{t=1}^{N} \log P(y_t \mid x, y_{<t}) \quad \textbf{(1)}$$

To prevent acoustic over-fitting, SpecAugment is applied during LoRA adaptation with time-masking (parameter T=50) and frequency-masking (parameter F=27), enabling robust phoneme recognition across variable background noise.

B. BGE-M3 Multilingual Semantic Representation & Pseudo-Centroid

Candidate sequence y is encoded into a 1024-dimensional normalized vector e_y via BGE-M3: e_y = DensePool(Transformer_BGE(y)) in R^{1024} with $\|e_y\|_2 = 1$. From the K-best beam search candidates $H_K = \{y^{(1)}, ..., y^{(K)}\}$, an acoustic pseudo-centroid e_bar is established from the top-M candidates:

$$e_bar = \frac{1}{M} \sum_{m=1}^{M} e_{y^{(m)}}, \quad e_bar_norm = \frac{e_bar}{\|e_bar\|_2} \quad \textbf{(2)}$$

The semantic alignment score is calculated as the cosine similarity:

$$S_{BGE}(y, e_bar) = e_y^T * e_bar_norm = \cos(e_y, e_bar) \quad \textbf{(3)}$$

In addition, BGE-M3 computes multi-vector token interactions to measure fine-grained keyword alignment, ensuring that crucial technical loanwords are not dropped during decoding.

C. 32-Dimensional Feature Space & XGBoost Dynamic Meta-Judge

For each candidate pair (x, y), a 32-dimensional feature vector z = [phi_ASR, phi_SEM, phi_CTX] in R^{32} is constructed:

• Acoustic Uncertainty (phi_ASR): Acoustic Log-Loss L_ASR, Perplexity exp(L_ASR), maximum token entropy max_t H(P_t), and token log-likelihood standard deviation.
• Semantic Consistency (phi_SEM): Dense cosine similarity S_BGE, semantic distance to top-1 candidate, and sparse lexical matching score.
• Signal & Context Dynamics (phi_CTX): Utterance duration (seconds), expected Code-Switching Ratio (rho_CS), Signal-to-Noise Ratio (SNR in dB), and RMS signal energy.

XGBoost minimizes the regularized logistic objective:

$$L_{XGB}(\theta) = \sum_{i=1}^{M} l(u_i, \hat{u}_i) + \sum_{k=1}^{T} \left[\gamma * T_k + 0.5 * \lambda * \sum_{j=1}^{T_k} w_{kj}^2 \right] \quad \textbf{(4)}$$

The instance-specific dynamic threshold is estimated in real time via:

$$T_{dynamic}(x) = \mu_T + \sigma_T * \tanh(w^T * z_{acoustic} + b) \quad \textbf{(5)}$$

D. Closed-Loop Re-Decoding Algorithm

A hypothesis y is accepted if J_XGB(z) >= T_dynamic(x). If rejected, a closed-loop re-decoding pass is executed with selective logit penalties delta_l = -alpha * (1 - S_BGE) applied to suspicious boundary tokens, shifting decoder mass toward valid alternate phonetic hypotheses.

V. EXPERIMENTAL SETUP

Table I outlines the benchmark corpora utilized throughout training and evaluation:

TABLE I: TRAINING AND EVALUATION DATASET DISTRIBUTION

Split / Target	Corpus Name	Size	Linguistic Domain & License
ASR LoRA Fine-tune	SEA Code-Switching Speech [14]	45.3 hrs	10,306 natural bilingual EN-VI utterances (Research)
ASR LoRA Fine-tune	ViMedCSS [15]	32.64 hrs	Medical and pharmaceutical terminology (CC BY 4.0)
ASR Pre-adaptation	GigaSpeech 2 (Subset) [16]	350.0 hrs	Multi-dialect conversational Vietnamese (Non-commercial)
ASR Benchmark & Test	ZAIC 2026 LifeMate [17]	6.83 hrs	5,036 real-world assistant & smart device queries (Research)
BGE-M3 Contrastive	mMARCO Vietnamese [18]	39.78M	Query-positive-negative passage triplets (Apache 2.0)
BGE-M3 Cross-lingual	PhoMT Bilingual [19]	3.02M	Parallel Vietnamese-English sentence pairs (CC BY-NC 4.0)
BGE-M3 Hard Negatives	UIT-ViQuAD 2.0 [20]	35,000+	Challenging unanswerable QA context pairs (Research)

Audio signals are resampled to 16 kHz mono. 80-channel log-mel filterbanks are computed with a 25 ms window and 10 ms hop size. LoRA adaptation is optimized using AdamW with beta_1=0.9, beta_2=0.98, weight decay 0.01, and linear warmup for the first 1000 steps. The XGBoost meta-judge is trained using 500 boosting rounds with early stopping on validation loss.

VI. RESULTS AND DISCUSSION

A. Main Benchmark Evaluation (Table II)

Table II presents comparative performance on the LifeMate Code-Switching Test Set (786 speaker-independent utterances):

TABLE II: MAIN BENCHMARK RESULTS ON LIFEMATE CS TEST SET

System Architecture	Params	WER (%)	MER (%)	E-WER (%)	Hallucination (%)	RTF
Whisper-Base Zero-shot [4]	74M	34.20%	41.50%	38.90%	1.85%	0.014
Whisper-Small LoRA Fine-tuned	244M	8.45%	10.15%	12.40%	0.62%	0.019
Whisper-Large-v3-Turbo Zero-shot	809M	13.45%	18.20%	19.40%	0.92%	0.038
Whisper-Large-v3-Turbo LoRA	845M	4.85%	5.90%	6.80%	0.28%	0.039
+ Static Cutoff (tau=0.85)	845M	6.80%	8.50%	9.80%	0.45%	0.042
Proposed (ASR + BGE-M3 +	**1.377B**	**2.85%**	**3.40%**	**4.15%**	**0.06%**	**0.0415**

XGBoost)

Key Finding: The proposed pipeline achieves a breakthrough Word Error Rate (WER) of 2.85%, surpassing the competitive target of WER < 3.0%. Crucially, English loanword errors (E-WER) drop from 19.40% to 4.15% (a 78.6% reduction). Real-Time Factor remains at RTF = 0.0415, ensuring sub-second response times.

B. Ablation Analysis (Table III)

Table III breaks down the empirical contribution of each individual module:

TABLE III: ABLATION STUDY OF ARCHITECTURAL MODULES

Configuration	WER (%)	Delta WER	Recovery Rate (%)	Mean Re-decodes
ASR LoRA Base	4.85%	Baseline	0.0%	0.00
+ BGE-M3 Reranking Only	4.10%	-0.75%	61.4%	0.00
+ Static Confidence Cutoff	6.80%	+1.95%	48.2%	0.32
+ XGBoost T_dyn (without BGE)	3.65%	-1.20%	74.5%	0.15
+ Full Proposed System	**2.85%**	**-2.00%**	**93.8%**	**0.12**

C. Comparison of 5 Alternative Model Pairs (< 1.5B) (Table IV)

Table IV compares five distinct ASR + Embedding model combinations strictly under 1.5B parameters:

TABLE IV: COMPARISON OF 5 ALTERNATIVE MODEL PAIRS (< 1.5B PARAMETERS)

No.	Model Pair (ASR + Embedding)	Params	WER (%)	E-WER (%)	RTF	VRAM (GB)
1	**Whisper-Turbo + BGE-M3 (Proposed Champion)**	**1.377B**	**2.85%**	**4.15%**	**0.0415**	**4.8 GB**
2	Whisper-Small + Multilingual-E5-Base (Edge)	0.522B	6.42%	9.15%	0.0185	2.1 GB
3	PhoWhisper-Small + Vietnamese-Bi-Encoder	0.354B	8.75%	15.30%	0.0162	1.6 GB
4	Whisper-Medium + Multilingual-E5-Large	1.329B	3.75%	5.20%	0.0520	5.4 GB
5	PhoWhisper-Medium + BGE-M3 (Hybrid)	1.337B	4.10%	7.45%	0.0510	5.2 GB

D. Hypothesis Testing & Statistical Significance (Table V)

Table V provides empirical hypothesis test outcomes across 786 independent test utterances:

TABLE V: STATISTICAL HYPOTHESIS TESTING OUTCOMES

Hypothesis	Evaluation Metric	Significance Level	Test Statistic	Scientific Outcome
H0	Overall WER (2.85% vs 13.45%)	p < 10^-15	t = 28.42	Decisively Rejected
H1	E-WER Reduction	p < 0.001	Welch's t = 19.86	Accepted (78.6%

	(4.15% vs 19.40%)			reduction)
H2	Hallucination F1 (0.938 vs 0.642)	$p < 0.0001$	Delta F1 = +0.296	Accepted (Superior to static)
H3	Budget 1.377B < 1.5B & RTF=0.0415	100% satisfied	Margin = 122.85M	Accepted (Real-time feasible)

VII. CONCLUSION

This work has presented a novel, sub-1.5 billion parameter architecture for Vietnamese-English code-switching speech recognition, integrating LoRA-adapted Whisper-Large-v3-Turbo, BGE-M3 cross-lingual semantic representations, and XGBoost dynamic thresholding. The proposed system establishes a new benchmark with a Word Error Rate (WER) of 2.85%, suppresses technical loanword errors by 78.6%, and eradicates catastrophic hallucinations (0.06%), all while maintaining real-time execution (RTF = 0.0415) on commercial single-GPU hardware. This research provides an effective, scientifically grounded solution for real-world bilingual conversational voice interfaces.

ACKNOWLEDGMENT

This research was supported by computational resources and benchmark evaluation data from the Zalo AI Challenge 2026 (LifeMate Track) and open-source multilingual corpora. The authors express sincere gratitude to colleagues at Quy Nhon University (QNU), University of Science and Technology of Hanoi (USTH), and Vietnam – Korea University of Information and Communication Technology (VKU), The University of Danang.

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